

2154. Keep Multiplying Found Values by Two

Difficulty: Easy

Link: <https://leetcode.com/problems/keep-multiplying-found-values-by-two/description/?envType=daily-question&envId=2025-11-19>

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Problem:

You are given an array of integers `nums`. You are also given an integer `original` which is the first number that needs to be searched for in `nums`.

You then do the following steps:

1. If `original` is found in `nums`, **multiply** it by two (i.e., set `original = 2 * original`).
2. Otherwise, **stop** the process.
3. **Repeat** this process with the new number as long as you keep finding the number.

Return the *final* value of `original`.

Example 1:

Input: `nums = [5,3,6,1,12]`, `original = 3`

Output: 24

Explanation:

- 3 is found in `nums`. 3 is multiplied by 2 to obtain 6.
- 6 is found in `nums`. 6 is multiplied by 2 to obtain 12.

- 12 is found in nums. 12 is multiplied by 2 to obtain 24.
- 24 is not found in nums. Thus, 24 is returned.

Example 2:

Input: nums = [2,7,9], original = 4

Output: 4

Explanation:

- 4 is not found in nums. Thus, 4 is returned.

Constraints:

- $1 \leq \text{nums.length} \leq 1000$
- $1 \leq \text{nums}[i], \text{original} \leq 1000$